Tayler Erbe

Applied AI & Data Scientist · Production LLM Systems · Agentic AI · MLOps

San Diego, CA | tayler.erbe@gmail.com | taylererbe.com | linkedin.com/in/tayler-erbe-194374141 | github.com/tayler-erbe

SUMMARY

Data scientist with four years owning end-to-end AI and analytics delivery in a regulated enterprise, functioning as the advanced analytics lead: production LLM pipelines, multi-stage ML prediction systems, GPU inference engineering and governance-ready project design. Rated Extraordinary/Distinguished and named an AITS Excellence Award Finalist. Outside work, founder of Skyrule, a personal project with a 15-agent AI engineering organization, a self-hosted vLLM stack and validator-gated LLM generation.

SKILLS

AI & LLMs: LLM pipelines, agentic AI, multi-agent orchestration, RAG, LLM evaluation, prompt engineering, vLLM, Ollama, Llama, Qwen, Claude Code

Machine learning: scikit-learn, gradient boosting, survival modeling, time-series forecasting, NLP, embeddings, FAISS vector search

Engineering & MLOps: Python, SQL, FastAPI, Linux, Azure App Service, Azure Blob Storage, Streamlit, pipeline monitoring, pytest

Governance: PII redaction (Presidio), privacy by design, statistical validation, executive and governance communication

EXPERIENCE

Data Scientist (Applied AI & Advanced Analytics) *Nov 2023 – Present*

AITS, University of Illinois System — AI Solutions Team *Remote*

- **Legislative intelligence platform:** built and delivered an Azure-hosted LLM and ML platform for policy analysts; the bill-passage model reached 75.1% precision and 76.2% recall with a median 21 days' warning on a held-out General Assembly, and introduction-time ROC-AUC rose from 0.561 to 0.845 after retraining.
- **GPU inference engineering:** deployed vLLM on air-gapped institutional hardware after diagnosing the exact network block, giving the institution continuous-batching GPU inference for the first time (5.01x batching capacity at smoke test).
- **Production recovery:** rebuilt a lost quarterly workforce-forecasting system end to end in Python after its platform was decommissioned, restoring the deliverable the same day and correcting long-standing statistical defects.
- **Technical leadership:** lead the archival AI program from proof of concept to production, unifying three proofs of concept into one eight-stage pipeline and mentoring the partner organization's engineer toward full ownership.
- **Leadership:** recruited, onboarded and led 7+ data science interns over three years; authored governance-ready project dossiers for the Enterprise Systems Governance Committee.

Data Engineer *May 2022 – Nov 2023*

AITS, University of Illinois System — Advanced Analytics Team *Illinois*

- Built Python ETL and analytics pipelines for payroll, HR and enrollment reporting; assumed the team lead role in early 2023.

Founder & Technical Lead (Personal Project) — Skyrule *Aug 2026 – Present*

Agentic AI relational-intelligence platform · skyrule.io *Self-hosted*

- **Agentic AI:** designed a 15-agent AI engineering organization with code-ownership guardrails enforced as pre-execution hooks, and an autonomous job queue and GPU scheduler that has run 650+ unattended jobs.
- **LLM validation:** built a validator-gated, human-in-the-loop generator; each new check is tested against founder-labelled defects (8 of 8 caught) and by mutation testing (21 of 21 caught).

- **Inference:** self-hosted vLLM with AWQ/FP8 Qwen3 models on an RTX 5090 at 14,200–15,100 pieces per GPU-hour.

EDUCATION

B.S. Cognitive Science — Machine Learning & Neural Computation *2021*

University of California, San Diego

RECOGNITION & PUBLICATIONS

- AITS Excellence Award Finalist (2023–2024)
- Technical writing on production ML monitoring, Azure deployment and GPU inference: medium.com/@tayler.erbe